box. Enter the Vercel AI SDK (https://vercel.com/docs/ai), an open source framework developed by Vercel that offers powerful abstractions to streamline the development of AI projects in JavaScript and TypeScript.

In this chapter, we will explore the Vercel AI SDK's key features and see how they provide a more flexible and scalable approach to building AI-powered applications. We will address the three key challenges, starting with provider abstraction and real-time streaming, to build a more robust and professional application. We'll start with the key components of the Vercel AI SDK, and then we'll see how to integrate it into our Astra conversational AI agent app.

3.1 Introducing the Vercel AI SDK

To effectively tackle these challenges, we need a strong and flexible framework that can seamlessly integrate with various large language model (LLM) providers, manage streaming responses, and handle state management efficiently. The Vercel AI SDK is designed to address these needs and more.

3.1.1 Key features and benefits

Table 3.1 outlines some of the key features offered by the Vercel AI SDK. This chapter focuses on the first two features highlighted in the list: *provider abstraction* and s*treaming responses.*

Table 3.1 Key features of the Vercel AI SDK

Feature	Description
Provider abstraction	Provides a unified interface for interacting with different AI providers, allowing seamless switching between providers without modifying the core application logic
Streaming responses	Offers a consistent streaming API that works across different AI providers, enabling real-time display of partial results as they become available
State management	Provides a centralized store for managing conversation history and ensuring synchronization between the UI client and the server code
React server components support	Supports rendering UI elements on the server and streaming them to the client, minimizing the bundle size and improving the UI experience
Generating structured data	Enables the generation of structured data formats that adhere to schemas
Embeddings	Provides utilities for working with embeddings, enabling advanced use cases like semantic search and clustering
React hooks	Offers React hooks (`useChat`, `useCompletion`, `useAssistant`) for simplified integration of chat, completion, and interactive assistant features into React applications

Provider abstraction is the SDK's primary solution to the vendor lock-in problem. Instead of writing code that makes direct API calls to a specific provider (e.g., OpenAI, Google Gemini), the SDK provides a unified interface. This allows you to write one set